Chapter 10: Mechanics of Options Markets

Part 1 — Sections 10.1 to 10.3

Learning Objectives

After studying this part, you should be able to:

- Understand the basic characteristics of options.
- Differentiate between call and put options.
- Distinguish between American and European options.
- Understand the four possible option positions.
- Calculate option payoffs at maturity.
- Identify different types of underlying assets for options.
- Understand how stock, currency, index, and futures options operate.

10.1 Types of Options

Introduction

Options are fundamentally different from forward and futures contracts.

A forward or futures contract creates an **obligation** for both parties.

An option creates a **right but not an obligation** for the holder.

Therefore:

- Futures/Forwards → Obligation
- Options → Choice

Because of this flexibility, the buyer of an option must pay an upfront amount known as the **option premium**.

Call Options

A **call option** gives its holder the right to:

> Buy an underlying asset at a predetermined price on or before a specified date.

Key Components

Component	Meaning
Strike Price (K)	Price at which asset can be purchased
Expiration Date	Last date on which option can be exercised
Premium	Cost of purchasing the option

Example: Long Call Position

Suppose:

- Current stock price = $98
- Strike price = $100
- Premium = $5 per share
- Contract size = 100 shares
- Expiration = 4 months

Initial investment:

$$5 \times 100 = 500$$

Scenario 1: Stock Price at Expiration = $115

Gain from exercising:

$$115 - 100 = 15$$

Profit per share:

$$15$$

Total exercise gain:

$$15 \times 100 = 1500$$

Net Profit:

$$1500 - 500 = 1000$$

Scenario 2: Stock Price at Expiration = $102

Exercise gain:

$$102 - 100 = 2$$

Net profit:

$$2 - 5 = -3$$

Even though total profit is negative, exercising is still optimal because:

- Exercise → Loss = $3
- Don't exercise → Loss = $5

Therefore:

> An in-the-money call should always be exercised at maturity.

Put Options

A **put option** gives its holder the right to:

> Sell an underlying asset at a predetermined price on or before a specified date.

The put buyer benefits from falling prices.

Example: Long Put Position

Suppose:

- Strike Price = $70
- Premium = $7

- Contract Size = 100 shares

Initial investment:

$$7 \times 100 = 700$$

Scenario: Stock Price at Expiration = $55

Gain from exercising:

$$70 - 55 = 15$$

Profit per share:

$$15$$

Total gain:

$$15 \times 100 = 1500$$

Net Profit:

$$1500 - 700 = 800$$

Scenario: Stock Price at Expiration > $70

Option expires worthless.

Loss:

$$700$$

which equals the premium paid.

American vs European Options

American Option

Can be exercised:

> Any time before expiration.

Features

- Greater flexibility
- Usually more valuable
- Most exchange-traded stock options are American

European Option

Can be exercised:

> Only on the expiration date.

Features

- Simpler to analyze mathematically
- Used extensively in pricing theory
- Many index options are European

Early Exercise

American options provide the possibility of exercising before maturity.

Hull notes that there are situations where early exercise becomes optimal.

These situations will be studied in later chapters.

Important Exam Concepts (10.1)

Call Buyer Expects

✓ Rising prices

Put Buyer Expects

✓ Falling prices

Maximum Loss of Option Buyer

$$\text{Premium Paid}$$

Maximum Gain

Position	Maximum Gain
Long Call	Unlimited
Long Put	Limited (when stock price falls to zero)

10.2 Option Positions

Every option contract has two sides.

Long Position

Investor buys the option.

Pays premium.

Obtains rights.

Short Position

Investor writes (sells) the option.

Receives premium.

Accepts future obligations.

Four Possible Option Positions

1. Long Call

- Buy call
- Bullish

Expectations

Stock price increases.

Risk

Limited

Reward

Unlimited

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2. Short Call

- Sell call
- Bearish or Neutral

Expectations

Stock remains below strike.

Risk

Potentially unlimited

Reward

Limited to premium received

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3. Long Put

- Buy put
- Bearish

Expectations

Stock price falls.

Risk

Limited

Reward

Large if stock collapses

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4. Short Put

- Sell put
- Bullish or Neutral

Expectations

Stock remains above strike.

Risk

Substantial downside

Reward

Limited to premium received

Option Payoffs

Hull emphasizes that:

> Payoff ignores the initial premium.

Premium is considered separately when calculating profit.

Let:

$$K = \text{Strike Price}$$

$$S_T = \text{Stock Price at Expiration}$$

Long Call Payoff

$$\max(S_T - K, 0)$$

Interpretation

If:

$$S_T > K$$

Exercise.

Otherwise:

$$0$$

Short Call Payoff

$$-\max(S_T - K, 0)$$

Equivalent form:

$$\min(K - S_T, 0)$$

Long Put Payoff

$$\max(K - S_T, 0)$$

Interpretation

Exercise only when:

$$K > S_T$$

Short Put Payoff

$$-\max(K - S_T, 0)$$

Equivalent form:

$$\min(S_T - K, 0)$$

Payoff Summary Table

Position	Payoff
Long Call	$max(S_T - K, 0)$
Short Call	$-max(S_T - K, 0)$
Long Put	$max(K - S_T, 0)$
Short Put	$-max(K - S_T, 0)$

Profit vs Payoff

Many students confuse these concepts.

Payoff

Value obtained at maturity.

Example:

$$\max(S_T - K, 0)$$

Profit

Payoff minus premium paid.

Example:

$$Profit = \max(S_T - K, 0) - Premium$$

Risk Characteristics

Option Buyer

Advantages

- Limited downside
- Large upside

Maximum Loss

Premium only

Option Writer

Advantages

Receives premium immediately.

Disadvantages

Potentially large future obligations.

Important Hull Observation

The writer's profit/loss is exactly the opposite of the buyer's profit/loss.

Whatever one side gains, the other side loses.

10.3 Underlying Assets

Options can be written on many different underlying assets.

Hull introduces four major categories:

1. Stock Options
2. Currency Options
3. Index Options
4. Futures Options

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1. Stock Options

Most option trading worldwide involves stocks.

Contract Size

One contract generally represents:

$$100 \text{ shares}$$

This size matches the traditional stock trading lot.

Characteristics

- Exchange traded
- Highly liquid
- American style in most cases

Uses

Hedging

Protect stock positions.

Speculation

Benefit from expected price movements.

Income Generation

Writing options to collect premiums.

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2. Foreign Currency Options

Currency options allow traders to manage exchange-rate risk.

Trading Venue

Mostly:

- Over-the-counter (OTC)

Some:

- Exchange traded

Typical Contract Size

Usually:

$$10,000$$

units of foreign currency.

For Japanese Yen:

$$1,000,000$$

units.

Uses

Importers

Protect against currency appreciation.

Exporters

Protect against currency depreciation.

Speculators

Bet on future exchange-rate movements.

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3. Index Options

These options are written on stock market indices.

Common Examples

S&P 500

SPX

S&P 100

OEX

Nasdaq-100

NDX

Dow Jones Index

DJX

Characteristics

Most are:

- European options

Exception:

- OEX is American

Contract Size

Usually:

$$100 \times \text{Index Value}$$

Cash Settlement

Unlike stock options:

No delivery of underlying shares occurs.

Settlement is entirely in cash.

Example

Strike:

$$980$$

Index Value:

$$992$$

Payoff:

$$(992 - 980) \times 100$$

$$= 1200$$

Cash amount paid to option holder:

$$\$1200$$

Advantages of Index Options

Diversification

Exposure to entire market rather than one stock.

Lower Company-Specific Risk

Performance depends on index.

Useful for Portfolio Hedging

Portfolio managers frequently use index options.

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4. Futures Options

Options may also be written on futures contracts.

Characteristics

Usually:

- American style

Expiration occurs shortly before the expiration of the underlying futures contract.

Call Option on Futures

Holder gains when:

$$F > K$$

Payoff:

$$F - K$$

Put Option on Futures

Holder gains when:

$$F < K$$

Payoff:

$$K - F$$

Comparison of Underlying Assets

Underlying Asset	Typical Settlement
Stock	Physical delivery
Currency	Physical delivery
Index	Cash settlement
Futures	Futures position created

Section Summary

Types of Options

Call

Right to buy.

Put

Right to sell.

Exercise Styles

American

Exercise anytime before expiration.

European

Exercise only at maturity.

Four Positions

1. Long Call
2. Short Call
3. Long Put
4. Short Put

Payoff Functions

Long Call:

$$\max(S_T - K, 0)$$

Long Put:

$$\max(K - S_T, 0)$$

Short Call:

$$-\max(S_T - K, 0)$$

Short Put:

$$-\max(K - S_T, 0)$$

Major Underlying Assets

- Stocks
- Currencies
- Indices
- Futures

Quick Revision Sheet

Call Buyer

Expects price rise.

Put Buyer

Expects price fall.

American Option

Exercise anytime before expiry.

European Option

Exercise only at expiry.

Maximum Buyer Loss

Premium paid.

Maximum Short Call Loss

Unlimited.

Long Call Payoff

$$\max(S_T - K, 0)$$

Long Put Payoff

$$\max(K - S_T, 0)$$

Stock Option Contract Size

100 shares.

Index Option Settlement

Cash settlement.

Futures Call Payoff

$$F - K$$

Futures Put Payoff

$$K - F$$

End of Part 1 (Sections 10.1–10.3)